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# ManifoldNeuS: Manifold-aware View Optimizability for Pose-Free Neural Surface Reconstruction

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## Abstract

*Jointly optimizing camera poses and object geometry from unposed images is a challenging task in neural surface reconstruction. Existing methods often suffer from pose drift and geometric distortion, stemming from the easy-view bias — uniform view optimization favors easy-to-optimize views with abundant texture and good overlap that dominate gradient updates, while hard-to-optimize counterparts with weak texture or limited overlap yet critical for geometric completeness are progressively marginalized. To address this, we propose ManifoldNeuS, a novel method that explicitly models and leverages per-view optimizability to guide pose-free neural surface reconstruction. Specifically, we introduce the manifold-aware view optimizability score (MaVOS), which jointly assesses immediate fitness (the ease of optimizing each view) and long-term coverage gain (the value of optimizing each view) over the view-coherent manifold. Building on the MaVOS, we further devise a reconstruction pipeline that incorporates the per-view optimizability as a state control signal to guide the joint optimization process through three key components: dynamic view scheduling, gated positional encoding, and anti-score loss weighting. Experimental results on the benchmark dataset demonstrate that ManifoldNeuS outperforms existing methods in terms of accurate pose estimation and high-quality reconstruction, achieving robust joint optimization without known camera poses.*

## 1. Introduction

Multi-view 3D reconstruction is a cornerstone problem in computer vision and graphics. Recently, remarkable progress has been achieved in 3D reconstruction using neural radiance fields (NeRFs) [42] and 3D Gaussian Splatting (3DGS) [29]. However, these methods typically assume access to accurate camera poses, usually estimated by Structure-from-Motion (SfM) pipelines such as COLMAP [49], which limits their practical implementation when pose estimates are noisy or unavailable. Consequently, jointly optimizing camera poses and scene geometry has attracted considerable attention.

Despite it being a severely ill-posed optimization problem, recent efforts have explored various methodologies to tackle this challenge. Methods based on 3DGS achieve remarkable rendering quality, yet their explicit, discrete representation fundamentally conflicts with the thin nature of surfaces, limiting their ability to recover intricate geometry [22, 69]. Moreover, 3DGS-based joint optimization relies on additional well-initialized point clouds, increasing sensitivity to camera parameters [18]. Methods based on Transformer frameworks leverage strong data-driven priors to regress geometry and poses directly, but generally require large-scale datasets with ground-truth poses for supervision [6]. In contrast, neural implicit methods enable continuous surface representations well-suited for high-fidelity reconstruction via enforcing multi-view consistency and differentiable volume rendering [17, 34].

While previous works have explored joint optimization of camera poses and geometry within implicit representation frameworks [2, 23, 33, 48], they typically optimize all views uniformly, leading to a critical yet underexplored challenge, which we term *easy-view bias*. This bias manifests in two intertwined dimensions that destabilize joint optimization: First, local gradient dominance causes highly overlapping views to produce disproportionately strong gradients early in the optimization. Consequently, these "easy views" quickly lock their camera poses while dominating geometry updates, starving weakly-textured regions of essential gradient signals. Second, and more fundamentally, global coverage neglect arises when optimization favors immediate photometric accuracy over long-term geometric completeness. Candidate views covering previously unobserved regions provide critical topological value that outweighs their short-term optimization difficulty. Marginalizing them forces pose drift, resulting in the reconstruction of fragments, rather than the lack of local details.

To this end, we propose ManifoldNeuS to break the easy-

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view bias by fundamentally modeling the optimizability of views in pose-free neural surface reconstruction. The core idea is that a view's optimizability, namely whether it is both easy and worthwhile to optimize, is determined by both its immediate fitness, *i.e.*, how effectively it accelerates convergence and refines existing geometry, and its long-term coverage gain, *i.e.*, how much it expands topological coverage and helps recover unknown geometry. Crucially, evaluating coverage gain requires modeling topological relationships among views, where Euclidean metrics are insufficient. To formalize this insight, we introduce the manifold-aware view optimizability score (MaVOS), a novel metric that unifies both local constraint strength and global topological coverage. The former quantifies immediate fitness through feature matching, while the latter measures long-term coverage gain via minimum distances over an embedded view graph. Building on MaVOS, we propose a MaVOS-guided reconstruction method to transform per-view optimizability into overall optimization stability, including view scheduling with dynamic refinement to guide progressive optimization from local to global, MaVOS-gated positional encoding to prevent premature high-frequency interference on poses by modulating frequency bands, and anti-score loss weighting to rebalance gradient contributions across views.

In summary, our main contributions are as follows:

1. We identify the easy-view bias in pose-free neural surface reconstruction: uniform view optimization favors easy-to-optimize views while overlooking those valuable topology-enhancing ones, leading to pose drift and geometric distortion.
2. We introduce a manifold-aware view optimizability score that unifies local constraint strength and global topological coverage to quantify immediate fitness and long-term topological gain.
3. We develop a MaVOS-guided reconstruction pipeline that transforms per-view optimizability into overall optimization stability, thereby enhancing 3D reconstruction quality from unposed views.

## 2. Related Work

**Scene representations for 3D reconstruction.** Advances in multi-view 3D reconstruction are driven by three paradigms: transformer-based, 3DGS-based, and neural implicit methods. Transformer-based methods [6, 31, 50, 53, 56, 57], like DUST3R [57] and VGGT [53], enable feed-forward and calibration-free reconstruction by directly regressing 3D structure and poses, but require extensive supervised training on pose-annotated data; in contrast, we recover geometry and poses from unposed images without pose supervision. 3DGS [29] provides an explicit representation for high-quality novel view synthesis, modeling scenes as anisotropic Gaussians optimized via a Gaussian rasterizer. Despite extensions [4, 13, 21, 22, 38, 67, 69] targeting surface reconstruction, its discrete and unstructured point representation is primarily optimized for appearance rather than geometric regularization, yielding noisy and fragmented surfaces. In unposed settings, 3DGS faces additional challenges due to its dependency on well-initialized point clouds. Most methods thus utilize external priors, such as off-the-shelf geometric predictors [9, 11, 15, 66] or depth estimators [18, 32] for initialization, yet are unreliable without ground-truth poses and typically restrict applicability to ordered data [25, 26, 28, 61]. Consequently, few works have demonstrated robust surface reconstruction from unposed images within the 3DGS framework.

Neural implicit methods, initiated by NeRF [42], model appearance and geometry as continuous functions. Subsequent works, like UNISURF [43], VolSDF [65], and NeuS [55], advance geometric modeling by incorporating implicit occupancy fields or signed distance functions (SDFs) via volume rendering. Among them, SDF-based methods [17, 34, 36, 47, 55, 59, 65, 68] extract the surface as the zero-level set of SDF using the Marching Cubes algorithm [37], producing stronger geometric fidelity [48]. In this work, we build our framework upon neural implicit representations, particularly NeuS, owing to its surface fidelity.

**Pose-free neural implicit representation.** To eliminate reliance on precomputed poses, recent work jointly optimizes camera poses and neural scene representations with poses as learnable parameters. Early methods [5, 12, 60, 63] handle only forward-facing scenes; subsequent works support complex trajectories but require accurate pose priors, either near-ground-truth initializations [7, 16, 35, 44, 62] or known distributions [10, 39, 40, 70, 71]. Recent attempts adopt identity initialization to remove priors, yet suffer from local minima, motivating geometric cues (*e.g.* monocular depth, optical flow) [3, 24, 54] or incremental SLAM-inspired incremental registration [41, 45, 58, 64, 72].

The aforementioned methods target novel view synthesis; neural surface reconstruction demands far stricter pose accuracy. Minor misalignments propagate to severe geometric distortion, as SDFs require highly consistent multi-view constraints to reach the correct zero level set. Methods like PoRF [2] and SC-NeuS [23] handle noisy poses yet rely on coarse pose estimates, leaving truly pose-free surface reconstruction under-explored. NoPose-NeuS [48] leverages monocular-depth-supervised feature consistency; ParaSurRe [33] employs cluster-wise incremental reconstruction with PnP-based SDF fusion. However, both adopt uniform training, treating all views equally regardless of per-view optimizability differences. We address this by quantifying per-view optimizability to guide pose-free neural surface reconstruction.

## 3. Method

Given a collection of unposed images, our goal is to recover camera poses and neural surface geometry simultaneously. To address easy-view bias induced by uniform optimization in pose-free neural surface reconstruction, we propose a comprehensive method centered on the manifold-aware view optimizability score (MaVOS) (Sec. 3.2), as illustrated in Fig. 1. MaVOS quantifies per-view optimizability via local constraint strength and, crucially, globally topological coverage measured on a view-coherent manifold approximated from multi-view geometric consistency. This manifold-aware design yields a balanced MaVOS that jointly accounts for the ease and value of optimizing each view. We then integrate MaVOS into the optimization process via three complementary components (Sec. 3.3), which transform per-view optimizability into a state-aware control signal within the NeuS framework, enabling stable and accurate joint optimization from unposed images.

### 3.1. Preliminaries

Our work is built on the SDF-based neural surface reconstruction, which implicitly represents scene geometry and appearance with the zero-set level of SDF fields and color fields. Given a 3D point $\mathbf{x}$, the implicit SDF field maps it to a signed distance $d(\mathbf{x})$ and a geometry feature $\ell(\mathbf{x})$. The appearance is modeled by a color field, which takes the spatial point $\mathbf{x}$, its view direction $\mathbf{v}$, the normal $\mathbf{n}(\mathbf{x})$ calculated by the gradient of $d(\mathbf{x})$, and the geometry feature $\ell(\mathbf{x})$ as inputs and outputs color $\mathbf{c}$:

$$(d(\mathbf{x}), \ell(\mathbf{x})) = F_g(\mathbf{x}), \quad \mathbf{c}(\mathbf{x}) = F_c(\mathbf{x}, \mathbf{v}, \mathbf{n}(\mathbf{x}), \ell(\mathbf{x})), \quad (1)$$

where $F_g$ and $F_c$ are approximated via multi-layer perceptron (MLP) networks. To compute the color of an image pixel, we approximate volume rendering [27] by numerical quadrature. Given $n$ points $\mathbf{x}_i$ sampled along its camera ray $\mathbf{r}$, the expected color of that pixel is given by:

$$\hat{\mathbf{c}} = \textstyle\sum_{i=1}^{n} T_i \alpha_i \mathbf{c}_i, \\ T_i = \textstyle\prod_{j=1}^{i-1}(1-\alpha_j),\ \alpha_i = 1 - \exp\left(-\int_{t_i}^{t_{i+1}} \rho(t)\, dt\right), \quad (2)$$

where $\mathbf{c}_i$ is the output color by Eq. 1, $T_i$ represents the accumulated transmittance, $\alpha_i$ denotes the discrete opacity, $\rho(t)$ is an opaque density function from NeuS [55].

To capture high-frequency details, positional encoding (PE) [51, 52] maps inputs into a higher-dimensional space using functions with $L$ frequency bases. However, in joint optimization, high-frequency PE complicates effective updates of low-frequency pose parameters during backpropagation. To address this, BARF [35] introduces a coarse-to-fine strategy that gradually activates PE frequency bases via a weight function $\varphi$. Given $L$ total frequency bases, the $l$-th PE frequency base in BARF is defined as:

$$\gamma_l(\mathbf{x}; \beta) = \varphi_l(\beta) \cdot \left[\sin(2^l \pi \mathbf{x}), \cos(2^l \pi \mathbf{x})\right], \quad (3)$$

where $\varphi_l(\beta) = \frac{1}{2}\left[1 - \cos(\mathrm{clamp}(\beta - l, 0, 1)\pi)\right]$, and $\beta \in [0, L-1]$ is a controllable parameter proportional to the optimization progress.

For the camera parameterization, in this work, we assume that camera intrinsics are available and focus on estimating the camera pose $[\mathbf{R}_i|\mathbf{t}_i]$ with a rotation $\mathbf{R}_i \in \mathbf{SO}(3)$ and a translation $\mathbf{t}_i \in \mathbb{R}^3$.

### 3.2. Manifold-Aware View Optimizability Score

To quantify the per-view optimizability, we convert the immediate fitness and long-term coverage gain into local constraint strength and global topological coverage.

We first evaluate the ease of optimizing a view through local constraint strength, which reflects its convergence speed based on multi-view geometric consistency. This is naturally captured by feature co-visibility: views with strong feature correspondences receive stronger geometric constraints during pose estimation, leading to faster convergence and higher optimization stability [49]. Given an image collection $\mathcal{V} = \{v_i\}_{i=1}^N$, we therefore construct a co-visibility-matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$, where each element $W_{ij}$ denotes the number of feature correspondences between $v_i$ and $v_j$. Notably, $\mathbf{W}$ purely measures feature-level co-visibility without geometric verification (*e.g.*, RANSAC) at this stage. The local constraint strength for view $v_i$ is then computed as:

$$\sigma_i = \sum_{j=1}^{N} W_{ij} \Big/ \Big(\sum_{i=1}^{N}\sum_{j=1}^{N} W_{ij}\Big). \quad (4)$$

A view with a larger $\sigma_i$ indicates stronger immediate geometric constraints, enabling faster convergence during early optimization stages while dominating gradient updates.

However, local constraint strength is inherently limited by its pairwise correspondence model, which ignores scene geometry and fails to distinguish globally redundant views from topologically critical ones. For instance, two views covering unobserved regions may appear equally valuable in image space, yet only the one connecting disconnected geometric components provides critical topological gain.

To resolve global coverage neglect, we introduce a manifold-aware topological coverage that identifies structurally critical views within the scene geometry. We perform spectral embedding on the view graph induced by the co-visibility matrix $\mathbf{W}$. Let degree matrix $\mathbf{D}$ with $D_{ii} = \sum_{j=1}^{N} W_{ij}$, we construct the symmetric normalized Laplacian $\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$. We then solve $\mathbf{L}_{\text{sym}}\mathbf{u}_k = \lambda_k \mathbf{u}_k$, where $\lambda_k$ is the $k$-th smallest eigenvalue, and obtain the spectral embedding $\mathbf{E} = [\mathbf{u}_2, \ldots, \mathbf{u}_{d+1}] \in \mathbb{R}^{N \times d}$, where $\mathbf{u}_1$ is discarded as the trivial constant mode. This embedding projects feature-level co-visibility onto a low-dimensional manifold where we measure a view's topological coverage $e_i$ through its normalized minimum

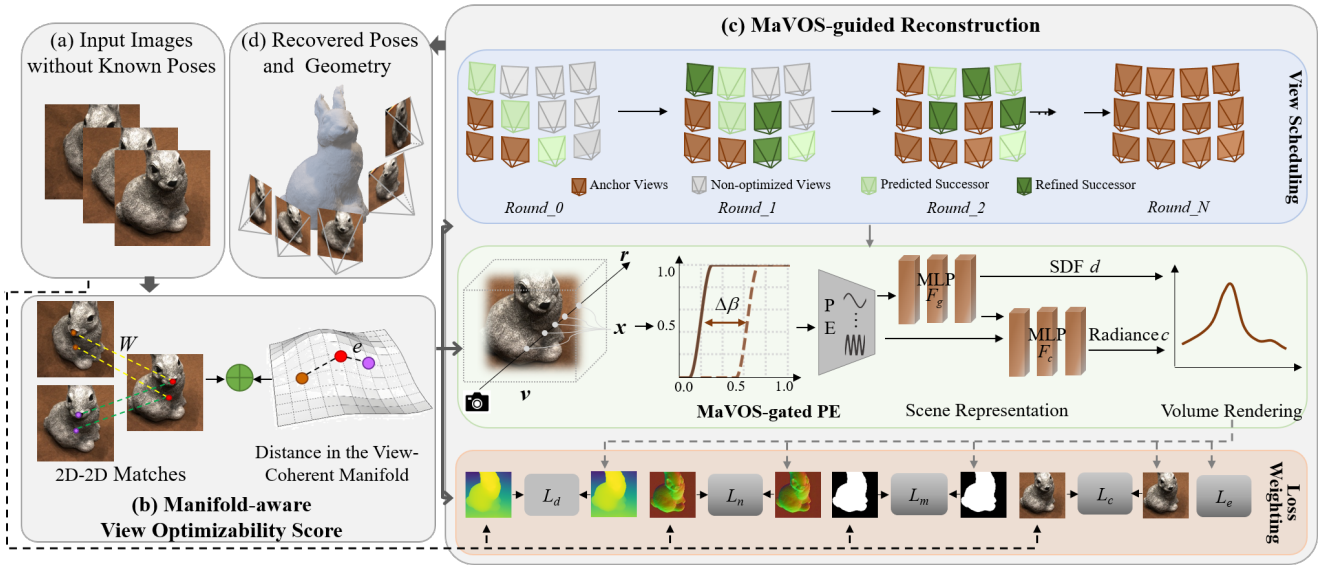



Figure 1. **Method Overview.** (a) Given multi-view image collections without known camera poses, (b) we compute a manifold-aware view optimizability score (MaVOS). MaVOS evaluates both the local constraint strength derived from feature matching and the global topological coverage measured by minimum geodesic distances on an embedded view graph. (c) Based on this, we design a MaVOS-guided reconstruction that progressively estimates camera poses and recovers scene geometry. Initially, we determine the optimization order of anchor views and their successors, dynamically reordering successor views after each update of the anchor set to enhance global consistency. For each optimization round, we introduce MaVOS-gated positional encoding that adaptively adjusts activation timing of high-frequency bands to prevent premature pose oscillation, and apply anti-score loss weighting to ensure balanced gradient contributions across all views. (d) By integrating these three components, we achieve a final 3D reconstruction with accurate camera poses and high-quality scene geometry.

distance to other views in the manifold space as:

$$e_i = \min_{v_j \in \mathcal{V}} \left\| \mathbf{E}_i - \mathbf{E}_j \right\|_2. \tag{5}$$

Larger $e_i$ values identify views in unexplored manifold regions that provide critical topological coverage.

By combing the local constraint strength $\sigma_i$ in Eq. 4 and the global topological coverage $e_i$ in Eq. 5, we obtain the unified view optimizability score $S_i$ as:

$$S_i = \alpha \cdot \sigma_i + (1 - \alpha) \cdot e_i, \tag{6}$$

where $\alpha \in [0, 1]$ serves as a balancing hyperparameter. Views with high $S_i$ values possess either strong multi-view constraints, significant potential to cover unexplored manifold coverage, or an optimal combination of both properties; conversely, low-scoring views lack both optimization stability and structural significance. This dual-aware scoring mechanism quantifies both the immediate fitness and long-term coverage gain of each view, providing principled guidance for the following reconstruction.

### 3.3. MaVOS-Guided Reconstruction

We apply this unified scoring to make the optimization view-aware within the NeuS framework. First, MaVOS-based view scheduling establishes a dynamic optimization sequence: we initialize reconstruction with a set of high-MaVOS views as anchors, then continuously reorder remaining views based on evolving MaVOS evaluations, enabling stable local-to-global convergence. Second, MaVOS-gated positional encoding adaptively modulates activation of frequency bases, specifically delaying high-frequency components for high-MaVOS views to prevent premature detail fitting from destabilizing pose estimation. Third, anti-score loss weighting calibrates gradient contributions by assigning higher weights to low-MaVOS views. Collectively, these components transform per-view optimizability into a state-aware optimization control signal for a stable joint optimization.

**View scheduling with dynamic refinement.** Our view scheduling strategy implements a progressive optimization paradigm: first establishing a geometrically stable core through reliable anchor views, then incrementally expanding reconstruction coverage through dynamically selecting remaining views.

We initialize reconstruction with the top-$K$ views ranked by MaVOS values to form the anchor set $\mathcal{A}$. The parameter $K$ is determined by the local constraint strength $\sigma_i$, as a high $\sigma_i$ value guarantees sufficient pairwise feature correspondences, which are essential for stabilizing initial pose and geometry estimation.

Following anchor initialization, we perform incremental

expansion through multiple optimization rounds. In each round, we select $M$ views from the candidate pool $\mathcal{R}$ using a greedy strategy that maximizes the refinement score:

$$S_{r_j} = \alpha \cdot \sigma_A(j) + (1-\alpha) \cdot e_A(j), \tag{7}$$

where $\sigma_A(j) = \frac{1}{|\mathcal{A}|}\sum_{v_i \in \mathcal{A}} W_{ij}$ quantifies the average covisibility between candidate view $v_j$ and the current anchors set $\mathcal{A}$, and $e_A(j)$ represents the normalized geodesic distance to its nearest anchor whin the manifold space. Given an initialized successor set as $\mathcal{M} = \varnothing$, the selection process iteratively identifies $v_j^* = \arg\max_{v_j \in \mathcal{R}} S_{r_j}$, appends it to $\mathcal{M}$, and dynamically updates the anchor set $\mathcal{A} \leftarrow \mathcal{A} \cup \{v_j^*\}$. This progressive expansion with refinement enables adaptive scene exploration with the exploitation of already-established structural constraints. Furthermore, in subsequent optimization stages, instead of identity initialization, we initialize each successor view's poses by inheriting the predicted camera poses of the anchor view with the most similar MaVOS value to reduce optimization oscillation and accelerate convergence by preserving pose continuity across stages.

**MaVOS-gated positional encoding.** Joint optimization of camera poses and scene geometry exhibits high sensitivity to PE activation schedules. While BARF [35] employs a global progress parameter for progressive frequency lifting, it homogenizes view-specific optimization characteristics. Our method introduces view-adaptive frequency gating through MaVOS-guided modulation that dynamically controls PE activation timing based on per-view optimizability.

Following the BARF [35], we first define the base frequency activation threshold $\beta_0$ for $L$ frequency bases at training progress $p \in [0, 1]$:

$$\beta_0 = [(p - p_a)/(p_b - p_a)] \cdot L, \tag{8}$$

where activation linearly progresses from 0 to $L$ over the scheduled interval $[p_\text{a}, p_\text{b}]$. Our key innovation introduces a MaVOS-driven offset $\Delta\beta_i$ that adapts this schedule per view:

$$\begin{gathered} \Delta\beta_i = \lambda \cdot \widetilde{S}_i \cdot \eta_f \cdot \eta_t, \\ \text{with } \widetilde{S}_i = S_i - \bar{S}, \quad \eta_f^{(l)} = l/L, \quad \eta_t = p^{\gamma_t}, \end{gathered} \tag{9}$$

where $\bar{S} = \frac{1}{|\mathcal{V}|}\sum_{v_j \in \mathcal{V}} S_j$ is the mean $S_i$ across view set $\mathcal{V}$ in the current optimization round, and $\lambda > 0$ scales the modulation magnitude. Instead of using $S_i$ directly, we use $\widetilde{S}_i$ to measure the view's relative optimizability, *i.e.*, deviation from the mean across the current view set to ensure robustness to global MaVOS distribution shifts during progressive reconstruction. $\eta_f$ applies delays to high-frequency components, preserving low-frequency geometric priors during early optimization. $\eta_t$ gradually increases modulation influence, accounting for initial pose instability and noisy MaVOS estimates in early training stages.

The final frequency scheduling combines base component in Eq. 8 and offset component in Eq. 9 as:

$$\beta_f = \beta_0 + \Delta\beta. \tag{10}$$

By introducing $\beta_f$ from Eq. 10 into Eq. 3, our method delays high-frequency activation for high-scoring views, preventing details from destabilizing the early-stage pose estimation process.

**Anti-score loss weighting.** While dynamic view scheduling and MaVOS-gated PE mitigate easy-view bias at the sampling and feature representation levels, they do not explicitly address gradient imbalance caused by uniform loss weighting across views. To directly rebalance supervision signals at the optimization level, we further employ an anti-score loss weighting strategy where each view's contribution to the total loss is weighted by the inverse score transformation $w_i = 1 - S_i$. Based on the anti-score weights, we jointly optimize parameters of camera poses and scene representation using the following loss function:

$$\mathcal{L} = \sum_{i=1}^{N} w_i(\lambda_c\mathcal{L}_c + \lambda_e\mathcal{L}_e + \lambda_m\mathcal{L}_m + \lambda_d\mathcal{L}_d + \lambda_n\mathcal{L}_n), \tag{11}$$

where $\lambda_*$ are balancing hyperparameters. For each input view $v_i$, we sample a set of $m$ rays and discretize each ray into $n$ points for supervision. The individual terms are defined as:

- **Photometric loss:** $\mathcal{L}_c = \sum \left\|\hat{C}(\mathbf{r}) - C(\mathbf{r})\right\|_2^2$, comparing rendered and ground-truth colors.
- **Eikonal loss** [20]: $\mathcal{L}_e = \sum (\|\nabla d(\mathbf{x}_{k,i})\|_2 - 1)^2$, regularizing the SDF gradient magnitude.
- **Mask loss:** $\mathcal{L}_m = \text{BCE}(M_k, \hat{O}_k)$, where $\hat{O}_k$ is the sum of weights along the camera ray $k$, and BCE denotes binary cross entropy.
- **Depth loss** [68]: $\mathcal{L}_d = \sum \left|\hat{\mathbf{D}}(\mathbf{r}) - \mathbf{D}^*(\mathbf{r})\right|$, with $\mathbf{D}^*$ being shifted and scaled depth [3, 46] and $\hat{\mathbf{D}}$ the rendered depth via volume rendering.
- **Normal loss:** $\mathcal{L}_n = \sum \left|\hat{\mathbf{n}}(\mathbf{r}) - \mathbf{n}(\mathbf{r})\right|$, aligning predicted normals with OMNI-DATA estimates [14].

# 4. Experiments

## 4.1. Experimental Setup

**Dataset.** To comprehensively validate the effectiveness of our method, we conduct experiments on the DTU dataset [1], which is widely used for evaluating 3D reconstruction methods. We chose the 9 scenes with a challenging variety of geometry, materials, and appearance for the experiments. Each scene contains 49 or 64 images. The monocular depth and normal supervisions are provided by MonoSDF [68].

**Compared Methods.** We compare our method against the following baselines: (1) NeuS-BARF, direct integration of

Table 1. Quantitative comparisons of camera pose estimation. We report the rotation angle error $\triangle R$ and the absolute translation error $\triangle T$ to evaluate pose accuracy. The term "--" denotes failed reconstructions. "Avg." indicates metrics averaged over all scenes, and the best results are highlighted in bold. NeuS-BARF and HT-3DGS cannot produce reliable poses, exhibiting large errors. SG-NeRF is initialized from noisy poses perturbed around ground-truth poses, yet these remain imperfect estimates. VGGT achieves competitive results on some scenes but fails on three. In contrast, our method significantly outperforms all state-of-the-art methods, demonstrating its effectiveness in accurate pose estimation from unposed image collections.

| Scan | NeuS-BARF [55] | | SG-NeRF [8] | | HT-3DGS [25] | | VGGT [53] | | Ours | |
|---|---|---|---|---|---|---|---|---|---|---|
| | $\triangle R\downarrow$ | $\triangle T\downarrow$ | $\triangle R\downarrow$ | $\triangle T\downarrow$ | $\triangle R\downarrow$ | $\triangle T\downarrow$ | $\triangle R\downarrow$ | $\triangle T\downarrow$ | $\triangle R\downarrow$ | $\triangle T\downarrow$ |
| 24 | 156.055 | 5.080 | 6.029 | 0.285 | 28.907 | 1.417 | **0.516** | 0.025 | 0.795 | **0.022** |
| 37 | 137.604 | 4.030 | 4.899 | 0.332 | 140.836 | 5.789 | 1.823 | 0.107 | **0.823** | **0.031** |
| 40 | 145.132 | 4.334 | 6.116 | 0.242 | 32.067 | 1.574 | 2.388 | 0.116 | **0.788** | **0.033** |
| 55 | 140.556 | 4.808 | 5.053 | 0.242 | 36.852 | 1.657 | 3.079 | 0.127 | **0.302** | **0.010** |
| 63 | 146.947 | 6.570 | 3.760 | 0.208 | 136.668 | 2.857 | -- | -- | **0.618** | **0.027** |
| 65 | 117.193 | 5.609 | 4.171 | 0.250 | 70.577 | 3.373 | 3.030 | 0.148 | **0.408** | **0.011** |
| 69 | 115.203 | 4.063 | 3.791 | 0.177 | 38.384 | 1.715 | 2.961 | 0.116 | **0.397** | **0.014** |
| 83 | 59.778 | 2.852 | 3.209 | 0.183 | 152.972 | 5.502 | -- | -- | **0.898** | **0.037** |
| 97 | 47.810 | 2.099 | 3.515 | 0.193 | 64.032 | 3.166 | -- | -- | **0.388** | **0.014** |
| Avg. | 118.475 | 4.383 | 4.505 | 0.235 | 101.320 | 4.224 | 2.300 | 0.106 | **0.602** | **0.022** |

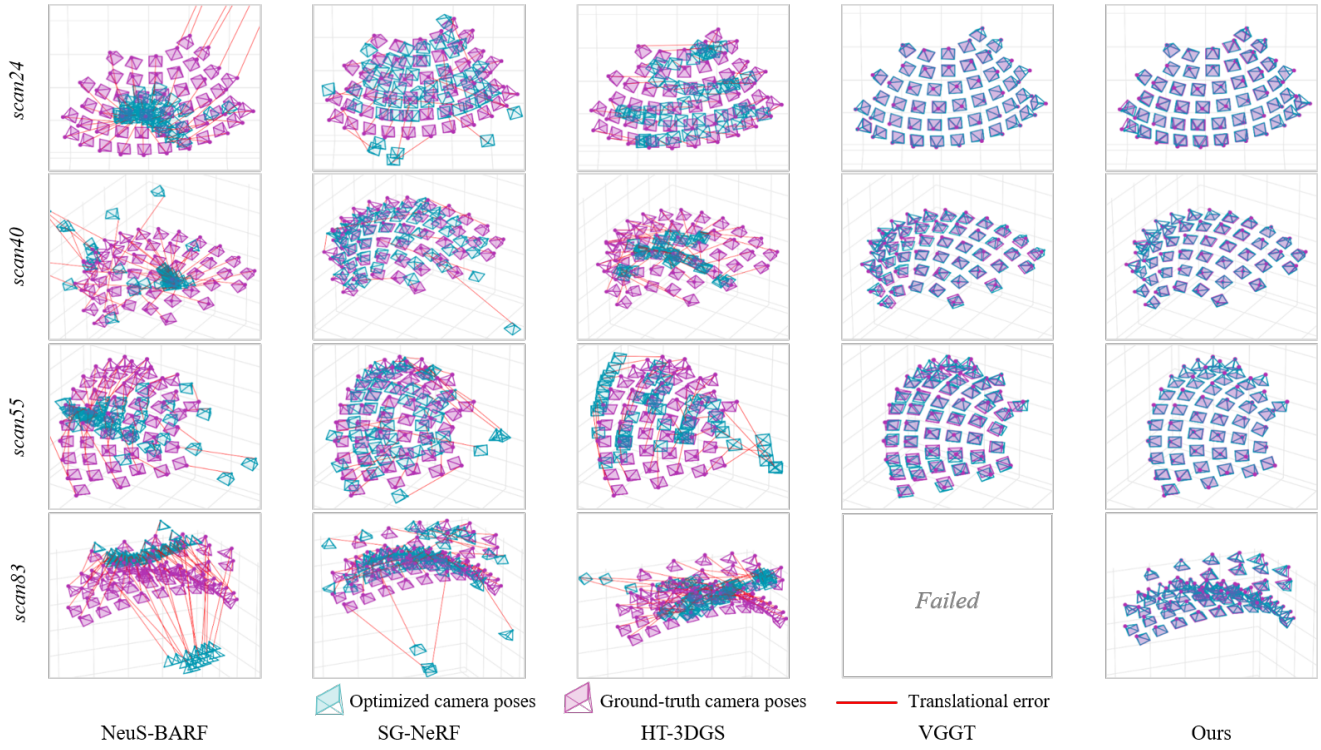



Figure 2. Qualitative comparisons of pose estimation. Our results closely match the ground-truth poses and exhibit fewer errors than the baseline methods.

BARF [35] into the NeuS framework for joint optimization; (2) SG-NeRF [8], a neural implicit method from noisy pose initializations perturbed around ground-truth; (3) HT-3DGS [25], a pose-free joint optimization method based on 3D Gaussian Splatting; (4) VGGT [53], a data-driven Transformer-based reconstruction method.

**Implementation details.** For the camera pose initialization, we provide SG-NeRF [8] with noisy poses for the initialization following their perturbation setting. The camera poses for all other methods are initialized as identity ma-

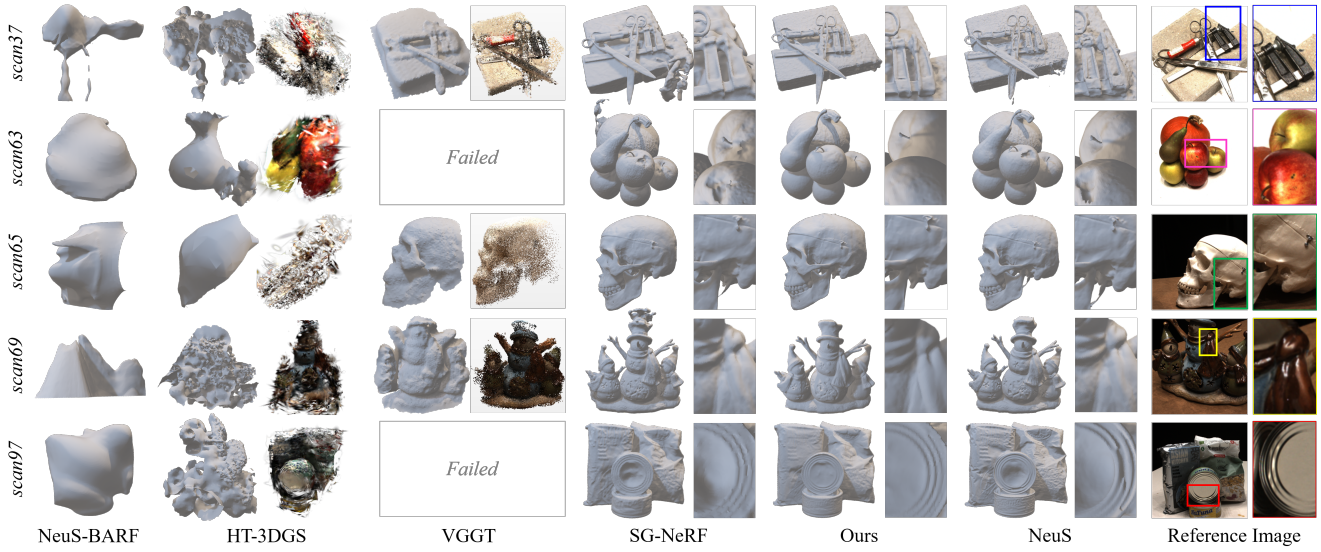



Figure 3. Qualitative comparisons of reconstruction quality. For HT-3DGS and VGGT, we show the corresponding SDF surfaces extracted from the output point clouds via Poisson reconstruction in the left column. We further provide their original representations on the right for better comparisons, *i.e.*, Gaussian primitives for HT-3DGS, and point clouds for VGGT. For other methods, we extract geometric meshes by marching cubes [37], with a resolution set to 512. One input image is shown in the last column for reference. Zoom-in views are given on the right of each column. NeuS-BARF and HT-3DGS produce severely distorted geometry due to inaccurate pose estimation. VGGT yields reasonable point clouds in some scenes. However, it suffers from failed reconstruction in *Scan63* and *Scan97* and produces incoherent surfaces. SG-NeRF reconstructs geometric artifacts and noise even with pose initialization close to the ground truth. In contrast, our method consistently delivers high reconstruction quality across diverse scenes, recovering fine geometric details on par with NeuS. Zoom in for details.

trices. The camera intrinsics are assumed to be known and provided by the dataset. All models are trained on a single NVIDIA RTX 5090 GPU. We use two separate Adam optimizers [30] to optimize camera poses and scene representation parameters. For pose estimation, we train our model $50K$ times per optimization round and randomly sample 256 rays per optimization step, with each ray consisting of 128 samples. After predicting all poses, we freeze the pose parameters and train the scene representation $200K$ times. For the first-round estimation of anchor views, all base training loss weights in Eq. 11 are set to 1.0. For the remaining optimization, except for the color loss, the other weights are reduced to 0.1. The balancing hyperparameter in Eq. 6 is set to 0.65. $K$ is set to cover $30\%$ of the total views. The scheduled interval $[p_{\text{a}}, p_{\text{b}}]$ is set to $[0.3, 0.7]$ for pose estimation, and $[0.1, 0.5]$ for geometry refinement. More details are provided in the supplementary material.

**Evaluation metrics.** We compute the rotation angle error $\triangle R$ and the absolute translation error $\triangle T$ after aligning the estimated poses with the ground truth using Procrustes analysis on the camera locations [19], following BARF [35]. The ground-truth poses are provided by the dataset. For 3D reconstruction, we use the Chamfer distance to measure the reconstruction quality.

### 4.2. Comparisons with the State-of-the-art Methods

We report quantitative results on camera pose accuracy in Table 1 and visualize them in Fig. 2. It can be observed that NeuS-BARF suffers catastrophic failure when poses diverge, indicating that naive adoption of BARF within neural implicit representations is ineffective. This highlights that SDF-based neural implicit methods face greater challenges in pose estimation than novel view synthesis, making it highly sensitive to initial pose inaccuracies and prone to optimization collapse. SG-NeuS is designed for noisy pose reconstruction where inputs are close to the ground truth. Remarkably, our method, initialized to the identity, achieves lower errors, demonstrating superior performance in pose-free surface reconstruction. HT-3DGS exhibits relatively large pose errors, as with most 3DGS-based pose-free methods, because it relies on inter-frame cues and is thus constrained to video-stream scenes. Fundamentally, 3DGS-based approaches require not only accurate poses but also a well-initialized point cloud, making them less suitable for more challenging pose-free settings. VGGT, a data-driven Transformer-based method, achieves competitive performance on some scenes but fails to reconstruct in three test cases. Especially in multi-scale scenes like *Scan83*, VGGT shows failed reconstruction results. In con-

trast, our method outperforms state-of-the-art methods by a large margin across all scenes. This superior performance is further validated in Fig. 2, where the camera poses recovered by our method closely match the ground-truth poses, providing a reliable solution for camera pose estimation.

We further present comparative results of reconstruction quality. As shown in Fig. 3, NeuS-BARF and HT-3DGS completely fail, producing distorted or fragmented geometry due to diverging into inaccurate camera poses. VGGT, though achieving low pose errors, represents scenes as point clouds, which are unsuitable for generating smooth SDF surfaces and yield incomplete geometry. SG-NeRF yields reasonable reconstructions but produces noisy geometry, such as in *Scan37* and *Scan63*. Our method consistently outperforms all others, achieving faithful and complete surface reconstruction. Notably, our method recovers finer geometric details on challenging structures, such as thin structures in *Scan37* and *Scan65*, and reflective surfaces in *Scan63*, *Scan69*, and *Scan97*, than NeuS trained with COLMAP-estimated camera poses. This superiority is attributed to the accuracy of camera pose estimation, which enables high-quality scene reconstruction. We also evaluate the scene reconstruction performance using the Chamfer distance. As shown in Table 2, our method achieves superior or competitive results compared to state-of-the-art methods such as VGGT and NeuS. Overall, our method can effectively optimize camera poses and recover scene geometry from unposed views.

Table 2. Quantitative comparisons of scene reconstruction in Chamfer distance. Due to the poor performance of NeuS-BARF and HT-3DGS, we only report the comparative results with SG-NeRF, VGGT, and NeuS (trained with COLMAP poses).

| Method | 24 | 37 | 40 | 55 | 63 | 65 | 69 | 83 | 97 | Avg. |
|---|---|---|---|---|---|---|---|---|---|---|
| SG-NeRF [8] | 0.605 | 0.630 | 0.519 | 0.521 | 0.537 | 0.508 | 0.703 | 0.717 | 0.567 | 0.590 |
| VGGT [53] | 0.317 | 0.395 | 0.487 | 0.498 | – | 0.424 | 0.331 | – | – | 0.409 |
| Ours | 0.342 | 0.372 | 0.415 | 0.337 | 0.554 | 0.470 | 0.356 | 0.398 | 0.387 | 0.404 |
| NeuS [55] | 0.377 | 0.353 | 0.325 | 0.309 | 0.556 | 0.442 | 0.293 | 0.325 | 0.330 | 0.368 |

### 4.3. Ablation Studies

We conduct ablation studies to validate the effectiveness of each component in our method. All experiments are performed on the *Scan24*. As shown in Table 3, BARF exhibits large drift under unknown poses due to the easy-view bias and thus has the highest error. Adding MaVOS-based view scheduling yields a significant improvement because anchors provide reliable early constraints and help suppress bias, leading to stable convergence. MaVOS-gated PE delays high-frequency activation for low-optimizability views and further reduces pose oscillation caused by premature details. Replacing base loss weighting with anti-score weighting prevents error propagation and improves robustness by redistributing gradients. Finally, enabling manifolds within MaVOS consistently yields more accurate pose estimates with a more reasonable view optimizability evaluation. Besides, dynamically reordering successors continuously maintains uniform coverage and connectivity. Taken together, these components contribute synergistically, delivering robust pose recovery and high-fidelity reconstruction.

Table 3. Ablation of key components on pose estimation. "BARF" is the base model that naively integrates BARF into the NeuS framework for joint optimization. "View sched." incorporates the MaVOS-based view scheduling strategy into "BARF" to validate progressive optimization. "Freq. gate" examines positional encoding by enabling/disabling MaVOS-gated frequency modulation. "Anti-score" compares the loss weighting scheme by contrasting anti-score weighting against the base weighting. 'Manifolds" evaluates the manifold-coverage term in MaVOS to assess its contribution to view optimizability calculation. "Dyn. refine." assesses the dynamic refinement on the view scheduling. We use the rotation angle error $\triangle R$ and the absolute translation error $\triangle T$ for pose estimation.

| BARF | View sched. | Freq. gate | Anti-score | Mani-folds | Dyn. refine. | $\triangle R\downarrow$ | $\triangle T\downarrow$ |
|---|---|---|---|---|---|---|---|
| ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | 168.05 | 2.71 |
| ✓ | ✓ | ✗ | ✗ | ✗ | ✗ | 36.02 | 1.78 |
| ✓ | ✓ | ✓ | ✗ | ✓ | ✓ | 2.09 | 0.09 |
| ✓ | ✓ | ✗ | ✓ | ✓ | ✓ | 1.80 | 0.07 |
| ✓ | ✓ | ✓ | ✓ | ✗ | ✓ | 1.38 | 0.06 |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✗ | 4.07 | 0.14 |
| ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | 0.80 | 0.02 |

More experimental results, detailed analyses, and ablation studies are detailed in the supplementary material.

## 5. Conclusion

We present ManifoldNeuS, a novel method for joint camera pose and geometry optimization from unposed images. By discovering and addressing the cause of pose drift and geometric distortion, *i.e.*, the easy-view-bias in uniform view optimization, we introduce the manifold-aware view optimizability score (MaVOS). This metric uniquely balances local constraint strength and global topological coverage to quantify how effectively each view advances optimization. Building on MaVOS, our reconstruction pipeline integrates three synergistic components, including view scheduling, positional encoding, and loss weighting. Our experiments show that ManifoldNeuS achieves state-of-the-art performance in both pose estimation and geometry reconstruction. We believe that our method encourages further development of pose-free neural surface reconstruction in the future by introducing view optimizability.

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